Solution

Step 1 Arrange dividend and the divisor in standard form.

$3x^3 - 2x^2 + 7x - 5$ (standard form of dividend)

$x + 3$ (standard form of divisor)

Write the coefficients of dividend in the first row. Put '0' for missing term(s).

3 −2 7 −5 (first row)

Step 2 Find out the zero of the divisor.

$x + 3 = 0$ implies $x = -3$

Step 3 Write the zero of divisor in front of dividend in the first row. Put '0' in the first column of second row.

−3	3	−2	7	−5	(first row)
	0				(second row)

Step 4 Complete the second row and third row as shown below.

−3	3		−2	7	−5	(first row)
	0	−3×3	−9	−3×−11	33	−3×40	−120	(second row)
	3		−11		40		−125	(third row)

All the entries except the last one in the third row are the coefficients of the quotient.

Then quotient is $3x^2 - 11x + 40$ and remainder is -125.

Example 3.36 Find the quotient and remainder when $(3x^3 - 4x^2 - 5)$ is divided by $(3x+1)$ using synthetic division.

Solution

Let $p(x) = 3x^3 - 4x^2 - 5$, $d(x) = (3x + 1)$

Standard form: $p(x) = 3x^3 - 4x^2 + 0x - 5$ and $d(x) = 3x + 1$

To find the zero of $3x+1$; put $3x + 1 = 0$ we get $3x = -1$, $x = -\frac{1}{3}$

$\frac{-1}{3}$	3	−4	0	−5	
	0	−1	$\frac{5}{3}$	$\frac{-5}{9}$	
	3	−5	$\frac{5}{3}$	$\frac{-50}{9}$	(remainder)

$$3x^3 - 4x^2 - 5 = \left(x + \frac{1}{3}\right)\left(3x^2 - 5x + \frac{5}{3}\right) - \frac{50}{9}$$

3 Algebra_Term1.indd 110 02/12/2022 21:39:52